THE VIRTUAL MEET EXPERIENCE

2018 - 2019

HS VIRTUAL CHALLENGE

REGIONAL QUALIFIERS' MEET

COMPUTER SCIENCE

DO NOT OPEN TEST UNTIL TOLD TO DO SO

2018-2019 Virtual Challenge Regional Qualifiers' Meet – Computer Science

Note: Correct responses are based on Java, **J2sdk v 1.8.x**, from Sun Microsystems, Inc. All provided code segments are intended to be syntactically correct, unless otherwise stated (i. e. `error` is an answer choice) and any necessary Java 2 Standard Packages have been imported. Ignore any typographical errors and assume any undefined variables are defined as used. **For all output statements, assume that the System class has been statically imported…** ***import static java.lang.System.*;***

QUESTION 1

Which of the following is equivalent to the expression $123_{16} + ABC_{16}$?

A. `3038` B. `3039` C. `3040` D. `3041` E. None are equivalent

QUESTION 2

What is output by the code to the right?

A. `4` B. `10`
C. `14` D. `3`
E. There is no output due to a runtime error.

```
out.println(5+5 * 2-1 / 2-1);
```

QUESTION 3

What is output by the code to the right($'s represent spaces)?

A. `$$$4.5` B. `4.5$$$`
C. `4.5000` D. `4.500000`
E. There is no output due to an error.

```
String s=String.format("%6s", 4.5);
out.println(s);
```

QUESTION 4

What is output by the code to the right?

A. `8.0` B. `8`
C. `7.0` D. `7`
E. There is no output due to an error.

```
double d=7.6;
out.println(d.round());
```

QUESTION 5

What is output by the code to the right?

A. `-4` B. `4`
C. `-1` D. `1`
E. There is no output due to a syntax error.

```
String h="aplus"+ " compsci";
char a=h.charAt(7);
char b=h.charAt(h.indexOf(a)+3);
out.println(a-b);
```

QUESTION 6

What is output by the code to the right?

A. `True` B. `true`
C. `False` D. `false`
E. None of the above.

```
boolean a=false;
boolean b=true;
boolean c= a && b || a;
out.println(c ^  b ^ a);
```

QUESTION 7

How many stars are ouput by the code to the right?

A. `9` B. `10`
C. `11` D. `12`
E. There is no output due to a runtime error

```
for(int y=0;y<10;y++)
  out.print('*');
out.println('*');
```

QUESTION 8

What is the output by the code to the right?

A. 156 B. 157

C. 158 D. 1256

E. 1258

```
int y=56;
if(y==56 || ++y==57)
  out.print(1);
if(y<10 && y==y++)
  out.print(2);
out.println(y);
```

QUESTION 9

What is the output by the code to the right+-?

A. 0 B. 91

C. 78 D. 66

E. There is no output due to a syntax error.

```
int sum=0;
for(int y=0;y<=12;y++)
  sum+=y;
out.println(sum);
```

QUESTION 10

What is the output by the code to the right ?

A. 6 17 B. 5 17

C. 6 21 D. 5 21

E. There is no output due to an error.

```
int[]k= {1,5,21,17,3,};
out.print(k.length);
out.print(" "+k[3]);
```

QUESTION 11

What is the output by the code to the right($'s represent spaces)?

A. 15 B. 7

C. 8 D. 1

E. There is no output due to an error.

```
int y1=7;
int y2=y1++;
int h=y1-y2;
out.println(h*y1+y2);
```

QUESTION 12

What is the output by the code to the right ?

A. 29.0 B. 29

C. 21.0 D. 21

E. There is no output due to an error

```
String f="1 2 3 4 5 6 7";
Scanner s=new Scanner(f);
int h=1;
while(s.hasNext())
  h+=s.nextDouble();
out.println(h);
```

QUESTION 13

What is the correct order of precedence for the operators to the right ?

A. II, IV, III, I

B. IV, I, II, III

C. II, I, III, IV

D. I, IV, III, II

E. I, IV, II, III

```
I. ()
II. instanceof
III. ?:
IV. new (Object creation)
```

QUESTION 14

What is the output by the code to the right ?

A. -1 B. -3

C. -2 D. 2

E. There is no output due to a runtime error

```
int[]i= {1,3,7,12,17,212};
int y=5;
int h=Arrays.binarySearch(i, y);
out.println(h);
```

QUESTION 15

What is the output by the code to the right ?

A. A very large number. B. `0`

C. `-4` D. `-2`

E. There is no output due to a runtime error.

```
long h=Integer.MAX_VALUE*2;
out.println(h);
```

QUESTION 16

What is the output by the code to the right ?

A. `1234` B. `[1, 2, 3, 4]`

C. Output cannot be predicted until runtime.

D. There is no output due to a compile error.

E. There is no output due to a runtime error.

```
int[]k= {1,2,3,4};
out.println(k);
```

QUESTION 17

What is the output by the code to the right ?

A. `0 big b0g of apples`

B. `0 big bag of 0pples`

C. `a big bag of apples`

D. `0 big b0g of 0pples`

E. There is no output due to an error.

```
String h="a big bag of apples";
h.replace("a", "0");
out.println(h);
```

QUESTION 18

What is output by the line marked `//1` in the code to the right?

A. `abc` B. `cba`

C. `879` D. `324`

E. There is no output due to an error.

QUESTION 19

What is output by the line marked `//2` in the code to the right?

A. `[abc, cba, 879]` B. `[cba, 324, 879]`

C. `[879, cba, abc]` D. `[879, 324, cba]`

E. There is no output due to a runtime error.

```
Stack<String>s=new Stack<String>();
s.push("abc");
s.push("cba");
s.push("324");
s.pop();
s.push("879");
out.println(s.peek()); //1
out.println(s); //2
```

QUESTION 20

What is the best case runtime of a quick sort?

A. `O(logn)` B. `O(n^2)` C. `O(n)` D. `O(1)` E. `O(nlogn)`

QUESTION 21

What is output by the code to the right ?

A. `511 255 127 63 31 15 7 3 1`

B. `511 255 127 63 31 15 7 3 1 0`

C. `255 127 63 31 15 7 3 1`

D. `255 127 63 31 15 7 3 1 0`

E. There is no output due to an error.

```
int y=511;
while(y>0)
 out.print(y/=2+" ");
```

QUESTION 22

Which of the following could replace **<1*>** in the code to the right ?

A. `String d`

B. `double d`

C. `Integer d`

D. `int d`

E. More than one of the above

QUESTION 23

Which of the following could replace **<2*>** in the code to the right ?

A. `salaried=true`

B. `super();`

C. `super(true);`

D. Nothing is needed.

E. More than one of the above.

QUESTION 24

What is the output by the following client code?

```
List <Employee> emplist ;
emplist = new ArrayList <Employee>();
emplist.add(new SalariedEmp(50000.00));
emplist.add(new BasicEmp(12.50));
for(int i = 0; i < emplist.size(); i++) {
        if(emplist.get(i).sal()) {
          emplist.get(i).promote(4000);
        }
        else {
          emplist.get(i).promote(2.50);
        }
}
out.println(emplist.get(1));
```

A. `hourly pay : 15.0  days : 103`

B. `hourly pay : 12.50  days : 2`

C. `hourly pay : 15.0  days : 3`

D. `hourly pay : 12.5  days : 102`

E. There is no output due to an error.

QUESTION 25

Which computer science concept(s) are demonstrated in the code above?

A. overriding

B. overloading

C. polymorphism

D. All of the above.

E. None of the above.

```
abstract class Employee
{
  private boolean salaried;
  public static int days;
  public Employee(boolean sal) {
     salaried = sal;
     days++;
  }
  public String toString() {
     return days + "";
  }
public abstract void promote(<1*>);
public boolean sal()
                  {return salaried;};
}
class SalariedEmp extends Employee
{
  private double salary;
  public SalariedEmp(double d) {
     <2*>;
     salary = d;
     days += 100;
  }
  public void promote(<1*>){
     salary += d;
  }
  public String toString() {
     return "salary : " + salary +
                  "  days : " + days;
  }
}
class BasicEmp extends Employee{
  private double hourPay;
  public BasicEmp(double hp) {
     super(false);
     hourPay = hp;
     days++;
  }
  public void promote(<1*>) {
     hourPay += d;
  }
  public String toString() {
     return "hourly pay : " + hourPay
                + "  days : " + days;
  }
}
```

QUESTION 26

What is output by the code to the right?

A. -1 B. 1

C. A number less than -1 D. A number greater than 1

E. There is no output due to an error

```
String h="the big sad";
String a="the great depression";
out.println(a.compareTo(h));
```

QUESTION 27

What is output by the code to the right?

A. 0 B. 86

C. D2 D. d2

E. 212

```
int y=212;
String s=Integer.toString(y,8);
y=Integer.parseInt(s,6);
s=Integer.toString(y,12);
y=Integer.parseInt(s,13);
out.println(Integer.toHexString(y));
```

QUESTION 28

What is output by the line marked //q28?

A. 360 B. 60

C. 120 D. 81

E. 30

QUESTION 29

What is output by the line marked //q29?

A. 6720 B. 60480

C. 20610 D. 3024

E. 15120

QUESTION 30

What is the value calculated by method recur to the right?

A. x!-y! B. y!-x!

C. y!/x! D. x!/y!

E. (y-x)!

```
static int recur(int y, int x) {
 if(y<=x)
    return 1;
 return y--*recur(y,x);
}

///////client code//////////////////
out.println(recur(6,3)); //q28
out.println(recur(9,4)); //q29
```

QUESTION 31

Which of the following methods is not contained in the Stack class?

A. pop B.push C.peek D.remove E. All of the previously mentioned are contained in the Stack class

QUESTION 32

Which of the following will not insert the value "8" if used to replace **<1*>** in the code to the right?

A. new Double(8).toString()

B. new Integer(8).toString()

C. "8" D. Integer.toString(8,9)

E. All of the above satisfy the given condition.

QUESTION 33

What is output by the code to the right?

A. 8 B. false

C. 312 D. true

E. There is no output due to an error.

```
Queue<String>q;
q=new LinkedList<String>();
q.add("345");
q.add("312");
q.add("212");
q.peek();
q.add(<1*>);
q.add("912");
q.poll();
out.println(q.remove("8"));
```

QUESTION 34

Which of the following data structures has O(1) runtime in best case and O(n) for worst case for removal and addition of an item?

A. Binary Search Tree
B. Queue
C. Stack
D. Hash Table
E. Red-Black Tree

QUESTION 35

What is output by the code to the right?

A. `n`
B. `7`
C. Output cannot be determined until runtime.
D. There is no output due to a compile error.
E. There is no output due to a runtime error.

```
String h=null;
h+=7;
out.println(h.charAt(0));
```

QUESTION 36

What is output by the code to the right ?

A. `true true`
B. `true false`
C. `false true`
D. `false false`
E. There is no output due to an error.

```
String s=Integer.toString(248,2);
out.print
         (s.matches("1+0+")+" ");
out.println(s.matches("\\w+"));
```

QUESTION 37

What could replace **`<1*>`** in the code to the right so that the code will compile without error?

A. `abstract`
B. `static`
C. `default`
D. More than one of the above.
E. None of the above.

```
interface I{
 public int ret();
 <1*> void show(int y) {
  out.println(y*y);
 }
}
```

QUESTION 38

How many bytes are in the primitive type char?

A. 1
B. 2
C. 4
D. 3
E. 8

QUESTION 39

What is output by the code segment to the right?

A. 0

B. 1

C. 2

D. 3

E. There is no output due to an error.

```
ArrayList<Integer>a;
a=new ArrayList<Integer>();
a.add(212);
a.add(7);
a.add(6);
a.add(4);
a.add(6);
a.add(4);
a.add(2);
a.add(4);
a.add(6);
a.remove(Collections.max(a));
int h=Collections.frequency(a, 6);
int v=Collections.frequency(a,h+1);
out.println(v);
```

QUESTION 40

What is 8-bit 2's compliment value of the following decimal number?

-54

Name _____ Final Score _____ _____

School _____ 2nd Grading _____ _____

Division: 1A 2A 3A 4A 5A 6A Grade: 9 10 11 12 1st Grading _____ _____
score initials

The Virtual Challenge Meets
Computer Science - Student Answer Sheet

1)	_____	21)	_____
2)	_____	22)	_____
3)	_____	23)	_____
4)	_____	24)	_____
5)	_____	25)	_____
6)	_____	26)	_____
7)	_____	27)	_____
8)	_____	28)	_____
9)	_____	29)	_____
10)	_____	30)	_____
11)	_____	31)	_____
12)	_____	32)	_____
13)	_____	33)	_____
14)	_____	34)	_____
15)	_____	35)	_____
16)	_____	36)	_____
17)	_____	37)	_____
18)	_____	38)	_____
19)	_____	39)	_____
20)	_____	40)	_____

Total Number Correct _____ x 6 = _____

Total Number Incorrect _____ x -2 = _____

Final Score (record here and top right) _____

2018-2019
Virtual Challenge
Regional Qualifiers'Meet
Computer Science - KEY

1)	B
2)	C
3)	A
4)	E
5)	A
6)	B
7)	C
8)	A
9)	C
10)	B
11)	A
12)	B
13)	E
14)	B
15)	D
16)	C
17)	C
18)	C
19)	A
20)	E
21)	E
22)	B
23)	C
24)	A
25)	A
26)	D
27)	B
28)	C
29)	E
30)	C
31)	D
32)	A
33)	D
34)	D
35)	A
36)	A
37)	D
38)	B
39)	D
40)	11001010

Note to Graders:

- All provided code segments are intended to be syntactically correct, unless otherwise stated (e.g. error is an answer). **Ignore any typographical errors**.
- Any necessary Standard Java 2 Packages are assumed to have been imported as needed.
- Assume any undefined (undeclared) variables have been defined as used.

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